The Phillips Curve and the AS Curve

Macroeconomics B
Lecture 5 (Chapter 18)

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Where We Are

In the previous lecture you built the demand side of the short-run model:

- **Goods market:** the IS curve links output to the real interest rate.
- **Monetary policy:** central banks set the short rate via a Taylor rule.
- **Aggregate demand:** $\boxed{Y - \bar{Y} = Z - a(\pi - \pi^*)}$ slopes downward in (Y, π) space.

But AD alone does not determine inflation. Lecture 4 told you how inflation affects demand through monetary policy; today explains how demand and output feed back into inflation.

Question for today: given output, what determines the rate of inflation? The chapter's answer is the **Phillips curve** and its modern reformulation as the **AS curve**.

Lecture 4 Recap: Two Points to Revisit

- **How can a nominal policy rate affect the real rate?**

  The Fisher equation links the nominal interest rate to the expected real interest rate:

  $$1 + r \ = \ \frac{1+i}{1+\pi^e} \qquad \Rightarrow \qquad r \approx i - \pi^e.$$

  So the real rate rises when the nominal rate rises *unless expected inflation rises one-for-one*:

  $$\Delta r \approx \Delta i - \Delta \pi^e.$$

  In our model, the central bank sets the short-term nominal policy rate:

  $$i^p = \bar{r} + \pi^e + h(\pi - \pi^*) + b(y - \bar{y}).$$

  The relevant nominal borrowing rate can differ from the policy rate: $i = i^p + \hat{\rho}$, where $\hat{\rho}$ captures spreads, risk premia, mortgage spreads, and other financing wedges.

  Combining the two:

  $$r \approx i^p - \pi^e + \hat{\rho} = \bar{r} + h(\pi - \pi^*) + b(y - \bar{y}) + \hat{\rho}.$$

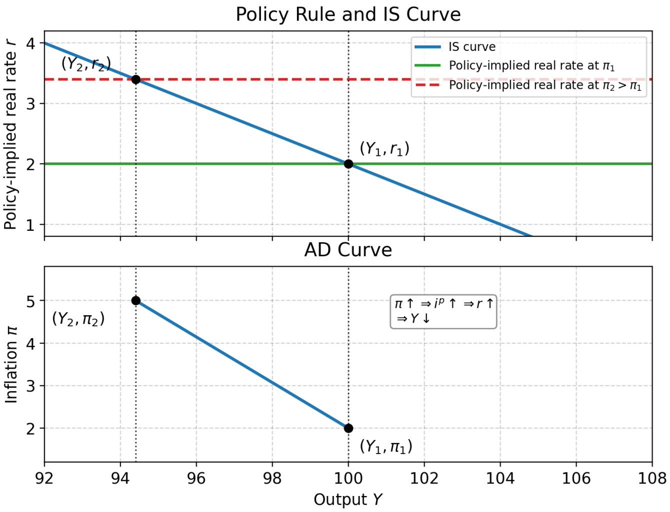
  Intuition: monetary policy moves demand because, holding expected inflation and spreads fixed, a higher nominal policy rate raises the real borrowing rate faced by households and firms.

Lecture 4 Recap: Two Points to Revisit

- **How do we get from policy rule plus IS to AD?**

  For each inflation rate π, monetary policy implies a real rate r. The IS curve then tells us the output level Y. Repeating this for different π values traces the AD curve in (Y, π) space.

Graphical Derivation: Policy Rule and IS to AD



The same output values Y_1 and Y_2 are projected from the policy-rule/IS panel to the AD panel. Higher inflation raises the policy-implied real rate and lowers output.

Roadmap for Today

1. **Lecture 4 recap: from policy rates to AD**
2. **Recent inflation: why we revisit the Phillips curve**
3. **Why prices and wages are sticky**
4. **The original Phillips curve and its 1970s breakdown**
5. **Expectations and the natural rate of unemployment**
6. **From the Phillips curve to the AS curve**
7. **Shocks, expectations, and the policy trade-off**
8. **Modern evidence: Denmark and the US since 2020**

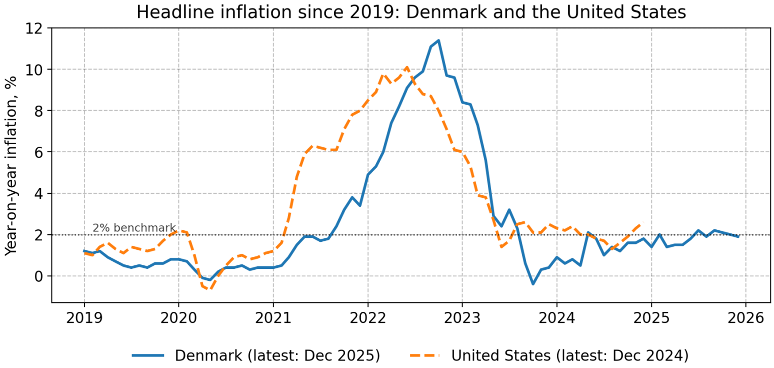
What You Will Be Able to Do After This Lecture

After this lecture, **you** will be able to:

- explain why nominal rigidities create a short-run link between activity and inflation,
- derive the **expectations-augmented Phillips curve** and the natural rate $\bar{u} = \mu/\alpha$,
- translate the unemployment Phillips curve into the AS curve $\pi = \pi^e + \gamma(y - \bar{y}) + s$ using an Okun-style link,
- distinguish movements *along* the AS curve from **shifts** due to π^e or supply shocks,
- interpret the recent Danish and US inflation episodes through the lens of the AS equation,
- compute the change in YoY inflation from monthly HICP data and read the slope of an empirical Phillips scatter.

Why Does This Matter? Recent Inflation in Denmark and the US

After two decades with inflation close to 2%, Denmark and the US saw a sharp inflation surge in 2021–2023, followed by rapid disinflation.



Headline HICP-comparable YoY inflation from Eurostat. Denmark peaks at 11.4% in October 2022; the US peaks at 10.1% in June 2022.

Inflation rose and fell quickly, but not all prices and wages adjusted at once.

This motivates the central question of the lecture:

Why do shocks move both inflation and real activity in the short run?

Motivation: What Price Stickiness Means

It means that many prices and wages are fixed for some time, or adjust only occasionally.

- Energy prices can move immediately.
- Retail prices may adjust with delays.
- Wages are often set in contracts and change less frequently.

Because firms and workers adjust prices and wages at different times, shocks temporarily change relative prices, markups, real wages, and demand.

If all prices and wages adjusted immediately, shocks would mostly show up in nominal variables. With sticky prices and wages, they also affect output and unemployment.

From Price Stickiness to the AS Curve

The recent inflation episode gives us three questions:

1. Why did inflation rise?

   $\rightarrow$ supply shocks and demand pressures

2. Why did output and unemployment move as well?

   $\rightarrow$ prices and wages adjust gradually

3. Why did inflation not immediately return to target?

   $\rightarrow$ expectations and delayed adjustment matter

These three ideas lead to the aggregate supply relation:

$$\boxed{\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t}$$

Inflation depends on expected inflation, economic slack, and supply shocks.

Why Are Prices Sticky? Menu Costs

Before deriving the Phillips curve, you need to understand **why** nominal adjustment is slow.

Menu costs (Mankiw 1985, Akerlof and Yellen 1985):

- changing prices has a fixed cost: reprinting catalogues, updating IT systems, renegotiating contracts;
- when the cost of changing prices exceeds the benefit, firms prefer to keep prices fixed;
- small menu costs at the firm level can generate large macroeconomic effects.

Empirical support: Nakamura and Steinsson (2008) find a median price duration of 8–11 months in US CPI data, excluding sales. Lecture 12 will revisit this with the **Calvo pricing** model.

Why Are Wages Sticky?

Staggered wage contracts (Taylor 1980):

- wages are renegotiated at fixed intervals, often every 1–3 years;
- not all contracts expire at the same time, so aggregate wages adjust gradually;
- in Denmark, many wages are shaped by collective bargaining between unions and employer organisations, with sector agreements and local negotiations transmitting shocks only with a delay.

Fairness and norms (Bewley 1999):

- managers report that nominal wage cuts damage morale and productivity;
- downward nominal wage rigidity is documented across countries.

Efficiency wages (Shapiro and Stiglitz 1984):

- firms may pay above market clearing to motivate workers and reduce shirking;
- this generates involuntary unemployment and slow wage adjustment.

From Micro Rigidities to the Phillips Curve

Sticky prices mean the price level adjusts gradually, so demand shocks move output away from $\bar{y}$ in the short run.

- When $y > \bar{y}$, firms produce more and hire more workers.
- The labour market tightens: vacancies are easier to open than to fill, so wages start to rise.
- Firms pass higher wage costs into their prices.

Together with inflation expectations, this gives a positive relation between activity and inflation:

$$\pi_t = \pi_t^e + \alpha(y_t - \bar{y}_t)$$

This is the modern Phillips curve in output-gap form. We now turn to its original formulation in terms of unemployment.

From the Output Gap to Unemployment

The output gap summarises the business cycle. **Unemployment** carries the same information from the labour market side.

- When $y > \bar{y}$, firms produce more and hire more, so u is low.
- When $y < \bar{y}$, firms cut back and shed labour, so u is high.

This tight co-movement is **Okun's law**. **Okun's law** states that output and unemployment move in opposite directions:

$$u_t - \bar{u} = -\lambda(y_t - \bar{y}), \qquad \lambda > 0.$$

We will use it later to translate between y and u, and between the Phillips curve and the AS curve.

Historically, the Phillips curve was first written in terms of u. Long unemployment series existed before reliable output-gap estimates, and wages are set directly in the labour market, so u is the proximate measure of wage pressure.

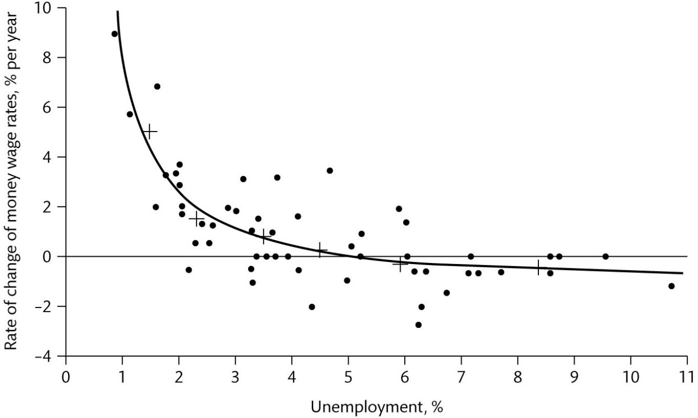
Phillips (1958): An Empirical Regularity

A. W. Phillips studied UK wage inflation and unemployment over a long historical sample.

- High unemployment: low or negative wage growth.
- Low unemployment: high wage growth.
- The relationship looked tight and **non-linear**: steep at low u, flat at high u.

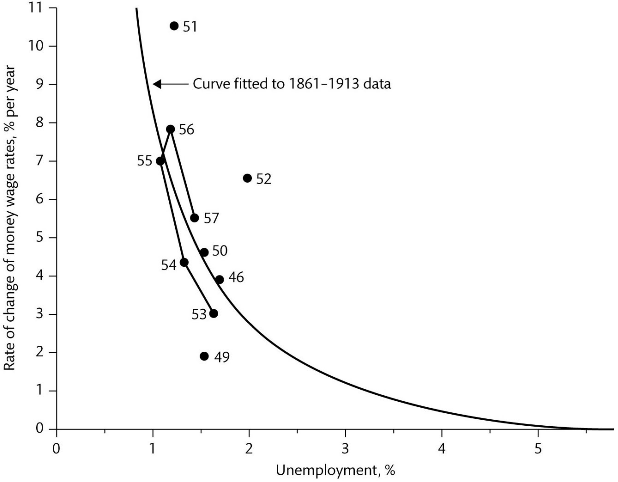
We first treat this as an empirical object, then add theory.

UK 1861–1913: The Original Phillips Curve



Source: Sørensen & Whitta-Jacobsen, Figure 18.1a.
The fitted curve is steep at low unemployment and flattens at high unemployment, consistent with tightening labour markets driving up wages faster.

The 1948–57 Period: Does the Curve Still Fit?



Source: Sørensen & Whitta-Jacobsen, Figure 18.1b.
Phillips fitted his curve to 1861–1913 data; the 1948–57 observations sit close to that same curve. The relationship looked stable across nearly a century.

A Simple Reduced-Form Phillips Curve

A stylised linear version of Phillips's relationship:

$$\boxed{\pi = \beta - \alpha\, u, \qquad \alpha > 0} \quad \text{(Book eq. 18.1)}$$

Sign intuition: low u means a tight labour market, wage pressure, and price pass-through. This is the mechanism from the previous section, now read off the labour market directly.

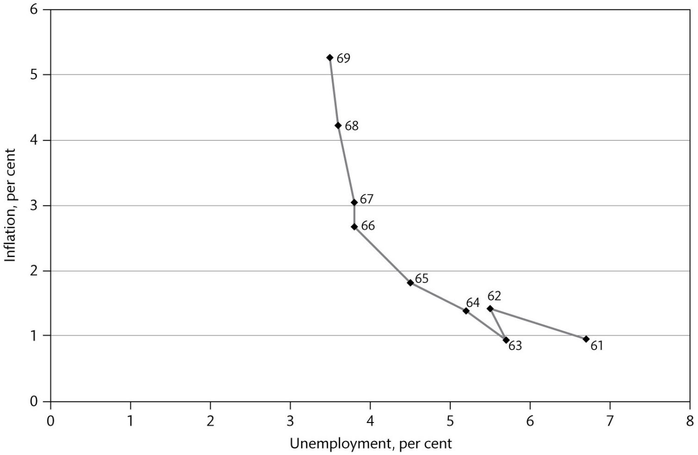
Parameters:

- α: slope, how strongly inflation responds to unemployment;
- β: intercept, shifts due to institutions, productivity, or pricing power.

Policy reading (Samuelson and Solow, 1960): The curve was interpreted as a stable *menu of choices*. A government wanting lower unemployment could pick a point further up the curve and accept higher π **permanently**.

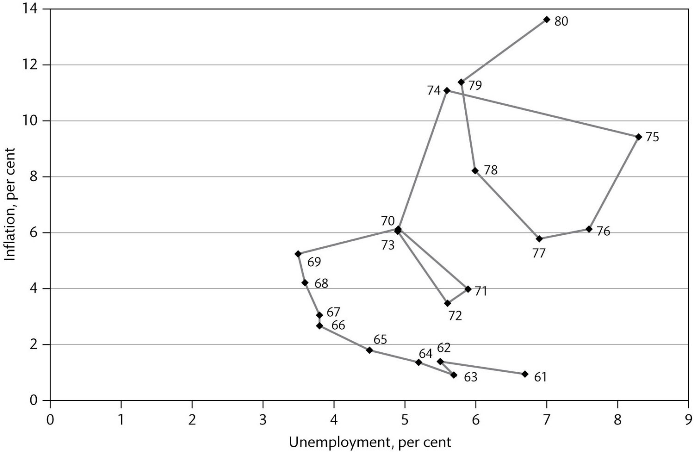
Limitation: Friedman (1968) and Phelps (1967) argued that workers and firms would eventually adjust their inflation expectations, so any trade-off must be short-run only.

The 1960s: Apparent Stability in the US



Source: Sørensen & Whitta-Jacobsen, Figure 18.2a.
US data 1961–69 traced out a clean Phillips trade-off, supporting policy prescriptions that targeted lower unemployment by tolerating higher inflation.

The 1970s: Stagflation Breaks the Trade-off



Source: Sørensen & Whitta-Jacobsen, Figure 18.2b.
Through the 1970s, the US (and most OECD economies) saw inflation and unemployment **rise together**: the simple curve no longer organises the data.

What Was Missing?

The breakdown is informative: it points to two missing ingredients in the simple curve.

- **Shifting expectations:** once inflation persisted in the late 1960s, agents started *expecting* higher inflation; that expectation worked itself into wage and price setting.
- **Supply shocks:** the 1973 and 1979 oil shocks raised costs at given output, pushing inflation up even with weak demand.

Lesson: the level of inflation cannot be explained by unemployment alone. **Expectations** and **shocks** matter.

Socrative Check: 1970s Breakdown

Go to **gosocrative.com** and enter room **MACROECONOMICSB**.

Question: In the 1970s, US inflation and unemployment rose at the same time. Which of the following is the *best* single-sentence reason why this is inconsistent with the simple Phillips curve $\pi = \beta - \alpha u$?

(A) The slope α became negative.

(B) Rising inflation expectations and supply shocks shifted β upward, raising π at any u.

(C) Unemployment was mismeasured during recessions.

(D) Phillips's UK curve was never relevant for the US.

(E) Loose monetary policy alone moved the economy along the curve to higher π and higher u simultaneously.

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Inflation Depends on Expected Inflation

If workers and firms expect higher inflation, nominal wage and price setting starts higher.
The expectations-augmented Phillips curve is:

$$\boxed{\pi \;=\; \pi^{e} \;+\; \mu \;-\; \alpha\, u} \qquad \text{(Book eq. 18.2)}$$

- π^e: expected inflation, formed before π is realised;
- μ: structural term capturing markups, bargaining power, regulations;
- $-\alpha u$: the original Phillips channel.

Expected inflation enters one-for-one because wage setters bargain over real wages: if prices are expected to rise by 3%, nominal wage demands start roughly 3% higher.

Where does μ come from?

The Natural Rate of Unemployment

Even in a well-functioning economy, some unemployment always exists: workers are between jobs, skills do not match vacancies, or firms hold wages above market-clearing to motivate workers. The unemployment rate consistent with these frictions, and with **no systematic upward or downward pressure on inflation**, is called the **natural rate** $\bar{u}$.

Formally, $\bar{u}$ is where $\pi = \pi^e$. Setting $\pi - \pi^e = 0$ in $\pi - \pi^e = \mu - \alpha u$:

$$\mu - \alpha \bar{u} = 0 \qquad \Rightarrow \qquad \boxed{\bar{u} = \frac{\mu}{\alpha}}$$

μ captures structural features: markups, bargaining power, benefit levels. Higher μ means higher $\bar{u}$.

Writing the Phillips curve in deviation form:

$$\pi - \pi^e = -\alpha (u - \bar{u})$$

- $u < \bar{u}$: labour market tighter than normal $\Rightarrow$ inflation above expectations.
- $u > \bar{u}$: labour market slack $\Rightarrow$ inflation below expectations.
- $u = \bar{u}$: inflation equals expectations and is stable. No long-run trade-off.

The Long-Run Phillips Curve is Vertical

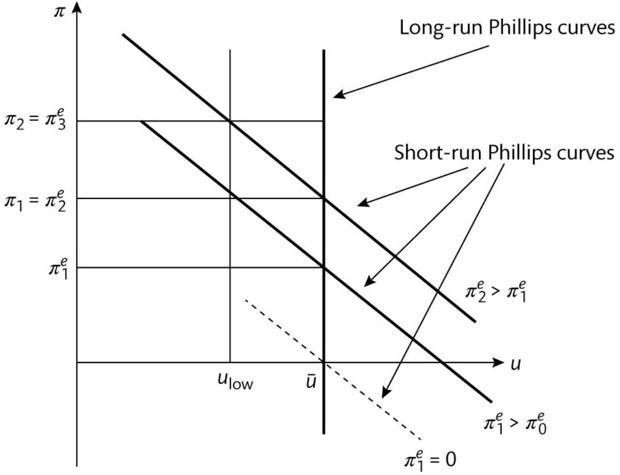
In long-run equilibrium $\pi = \pi^e$, so:

$$\pi - \pi^e \;=\; -\alpha\,(u - \bar{u}) \;=\; 0 \quad \Rightarrow \quad u = \bar{u}$$

regardless of the level of inflation.

- The long-run Phillips curve is **vertical** at $\bar{u}$.
- $\bar{u}$ is sometimes called the **NAIRU** (non-accelerating inflation rate of unemployment).
- Structural policies move $\bar{u}$, demand policies do not.

Multiple Short-Run Phillips Curves



Source: Sørensen & Whitta-Jacobsen, Figure 18.4.
Each downward-sloping curve corresponds to a different level of π^e. When expectations rise, the whole short-run curve shifts up; the long-run Phillips curve is vertical at $\bar{u}$.

Accelerationist Intuition

Why backward-looking? The simplest case: workers and firms expect next period's inflation to equal last period's. If inflation was 3% last year, they set wages and prices assuming 3% again. This is a reasonable rule of thumb when inflation has been stable for a while.

Substituting $\pi_t^e = \pi_{t-1}$ into $\pi_t - \pi_t^e = -\alpha(u_t - \bar{u}) + s_t$:

$$\boxed{\pi_t - \pi_{t-1} \; = \; -\alpha\left(u_t - \bar{u}\right) \; + \; s_t}$$

Why "accelerating"? If $u_t < \bar{u}$ every period, inflation rises by $\alpha(u_t - \bar{u})$ each period. It does not just stay high — it keeps climbing. $\bar{u}$ is therefore the rate at which inflation is *neither rising nor falling*: the **NAIRU**.

- Sustained $u_t < \bar{u}$: inflation accelerates upward.
- Sustained $u_t > \bar{u}$: inflation decelerates.
- $u_t = \bar{u}$: $\Delta\pi_t = 0$, inflation is stable.

Full derivation

From Unemployment to Output: An Okun-Style Link

The macro model works in (y, π) space, so we need a link from u to y. A reduced-form Okun relation:

$$\boxed{u_t - \bar{u} \;\approx\; -\lambda\,(y_t - \bar{y}), \qquad \lambda > 0}$$

- When output rises above potential, firms hire more, and unemployment falls.
- λ measures how strongly cyclical output translates into cyclical unemployment.
- For most advanced economies, λ is around 0.3–0.5.

This is taken as a given empirical regularity, not derived from a labour market model.

Deriving the AS Curve

Start from the deviation form of the expectations-augmented Phillips curve:

$$\pi - \pi^e \;=\; -\alpha\,(u - \bar{u}) \;+\; s$$

Insert the Okun link $u - \bar{u} = -\lambda\,(y - \bar{y})$:

$$\pi - \pi^e \;=\; \alpha\lambda\,(y - \bar{y}) + s$$

Define $\gamma \equiv \alpha\lambda > 0$:

$$\boxed{\pi_t \;=\; \pi_t^e \;+\; \gamma\,(y_t - \bar{y}) \;+\; s_t} \qquad \text{(AS curve)}$$

Why does $y > \bar{y}$ raise inflation? Follow the chain:

$y > \bar{y} \;\longrightarrow\; u < \bar{u} \;\longrightarrow\;$ firms hire more, even less productive workers $\;\longrightarrow\;$ marginal costs $\uparrow \longrightarrow\; \pi \uparrow$

Firms also face rising wages when the labour market is tight. Both channels push costs up and get passed into prices.

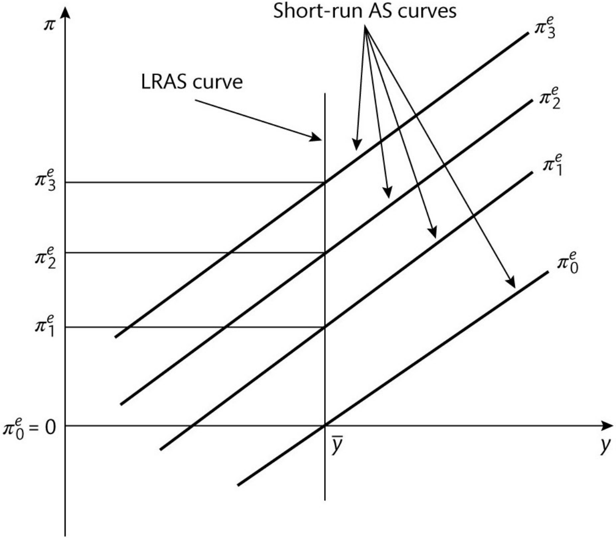
Graphical AS Intuition

The AS curve in (y, π) space:

- slopes **upward**: higher output raises marginal cost and pushes inflation up;
- passes through $(\bar{y}, \pi^e)$ at the natural level of output;
- shifts **up** when π^e rises or when a positive supply shock s hits;
- has a long-run counterpart, the LRAS curve, that is **vertical** at $\bar{y}$ once expectations have adjusted.

The slope γ matters for stabilisation policy: a flatter AS curve means inflation responds little to slack, so disinflating is costly in output terms.

Aggregate Supply Curves: SR and LR



Source: Sørensen & Whitta-Jacobsen, Figure 18.7.
Each upward-sloping line is a short-run AS curve at a given π^e. The vertical line at $\bar{y}$ is the long-run AS (LRAS) curve, which holds when expectations have fully adjusted.

Two Types of Shocks: Demand vs Supply

In AS-AD language:

- **Demand shock:** shifts AD, moves (y, π) *along* a given AS curve. Output and inflation move **together**.
- **Supply shock:** shifts AS, creates a *trade-off* between y and π. Output and inflation move in **opposite directions**.

Implication: after a supply shock, stabilising inflation requires a recession; after a demand shock, the central bank can stabilise both.

Socrative Check: Movement or Shift?

Go to **gosocrative.com** and enter room **MACROECONOMICSB**.

Question: Energy prices rise sharply, while expected inflation is still anchored. In the AS-AD diagram, what is the direct effect?

(A) A movement along the AS curve.

(B) An upward shift of the AS curve.

(C) A rightward shift of the AD curve.

(D) A movement along the AD curve only.

(E) A leftward shift of the AD curve.

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Supply Shocks in the AS Equation

Include s_t explicitly:

$$\pi_t \;=\; \pi_t^e \;+\; \gamma\,(y_t - \bar{y}) \;+\; s_t$$

Interpretation:

- $s_t > 0$: AS shifts up. Examples: oil price spike, increase in markups, fall in productivity.
- $s_t < 0$: AS shifts down. Examples: technology boom, terms-of-trade improvement.

The 2021–23 episode you saw at the beginning of this lecture likely combined large positive s_t (energy, supply chains) with positive demand shocks (post-COVID rebound, fiscal stimulus).

The Policy Trade-off after a Supply Shock

With $s_t > 0$, to keep inflation at target you typically need $y_t < \bar{y}$ for a while. From the AS equation, holding π^e fixed:

$$\pi_t - \pi_t^e \;=\; \gamma\,(y_t - \bar{y}) \;+\; s_t$$

Setting $\pi_t = \pi^*$ and $\pi^e = \pi^*$:

$$y_t - \bar{y} \;=\; -\frac{s_t}{\gamma}$$

- Larger shock s_t: larger required output gap.
- Smaller slope γ (flatter AS curve): larger output cost.

Sacrifice ratio intuition: how many percentage points of output must the economy give up to reduce inflation by one percentage point? A flat AS curve (γ small) means inflation barely responds to slack, so the central bank must engineer a large and prolonged recession to bring inflation down.

Sacrifice ratio details

Inflation Expectations: Static, Adaptive, Anchored

The behaviour of the AS curve depends critically on **how** expectations are formed:

- **Static:** $\pi_t^e = \pi_{t-1}$. Today's inflation moves directly into next period's expectations.
- **Adaptive:** $\pi_t^e = \pi_{t-1}^e + \eta(\pi_{t-1} - \pi_{t-1}^e)$ for $0 < \eta \leq 1$. Slow learning.
- **Anchored:** $\pi_t^e \approx \pi^*$ when the central bank's target is credible.

When expectations are **anchored**, the AS curve barely shifts after a temporary shock; the disinflation cost is small. When expectations are **static**, today's inflation persists into tomorrow.

Adaptive updating details

Why Unanchored Expectations Are Costly

Start from the AS curve:

$$\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t.$$

- If expectations are **anchored**, a temporary energy shock raises inflation today, but π_{t+1}^e stays close to target.
- If expectations become **unanchored**, the shock raises π_t, then higher realised inflation raises π_{t+1}^e.
- The AS curve keeps shifting up, so inflation persists even after the original shock fades.
- To bring inflation back, the central bank needs a larger or longer negative output gap.

$$\boxed{\text{unanchored expectations turn a temporary shock into persistent inflation}}$$

Group Exercise: Static vs Anchored Expectations

15 minutes, groups of 2–3. Open `L5_phillips_curve.ipynb`, section 4.

1. **Persistence.** Run the AS simulation with the original parameters. Then set `par['y_gap'] = -2.0`. Compare static and anchored expectations. Which regime creates more inflation persistence, and why?
2. **Output cost.** Keep `shock = 3.0` and `gamma = 0.4`. Find the negative output gap that returns impact inflation to target under anchored expectations. Then repeat with `gamma = 0.8`. What changes when the AS curve is steeper?

To Do: Fill in one row per group in the Response Sheet (Click the Link!) ; one sentence per task, plus the parameter value you changed.

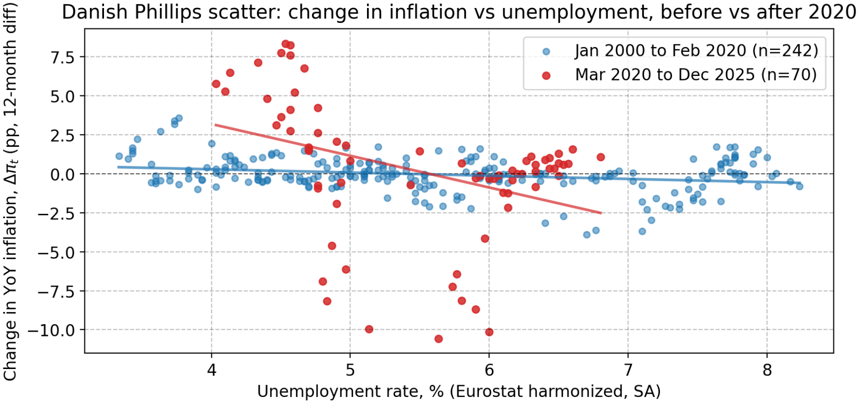
Modern Empirical Themes

Common findings in the empirical literature on the Phillips curve since the 1990s:

- the slope γ appears **flatter** in many advanced-economy samples;
- **anchoring** of expectations weakens pass-through from slack to inflation;
- **large supply shocks** can dominate demand conditions for years at a time;
- the Phillips relationship is **non-linear**: it may be steeper at very low u.

You will look at a Danish scatter next.

A Flatter Phillips Curve in Recent Data: Denmark Since 2000



Danish HICP inflation and unemployment from Eurostat. Pre-2020: $\hat{\kappa} = -0.21$, near-flat relationship. Post-2020: large vertical dispersion at moderate u, fingerprint of dominant supply shocks.

Real-World Example: Denmark Inflation 2021–2025

Danish HICP inflation rose from below 1% in early 2021 to a peak of **11.4%** in October 2022, then fell back to about 2% by mid-2024.

Through the lens of $\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t$:

- **Supply channel:** energy prices, gas, electricity, food (large positive s_t).
- **Demand channel:** post-COVID rebound and fiscal support raised $y - \bar{y}$ moderately.
- **Expectations:** the DKK peg to the euro and the ECB's anchor kept π^e broadly stable.

Class question: Why did Denmark's inflation come down so quickly compared with the 1970s episode?

Real-World Example: US Disinflation 2022–2024

US inflation fell from about 10% in mid-2022 to near 2% by 2024, with unemployment staying near 4%, no recession.

The model offers two readings:

- supply shocks were **transitory**: as s_t faded, the AS curve shifted back down;
- expectations remained **anchored**: the Fed's credibility kept $\pi^e \approx \pi^*$, so the disinflation did not require a deep recession.

This is the **soft landing** debate: was it luck (fading s_t) or skill (anchored π^e)? The interpretation requires that expectations did not drift upward persistently. **Class question:** Which of the two readings does the rapid disinflation in Denmark also support?

Takeaways

- Nominal rigidities (menu costs, sticky wages, efficiency wages) are the **micro foundation** for the Phillips curve.
- The simple Phillips curve $\pi = \beta - \alpha u$ broke in the 1970s because expectations and supply shocks were missing.
- The expectations-augmented version, $\pi = \pi^e + \mu - \alpha u$, removes the long-run trade-off; the long-run curve is vertical at $\bar{u} = \mu/\alpha$.
- Translating u into y via Okun gives the **AS curve**: $\pi = \pi^e + \gamma(y - \bar{y}) + s$.
- **Supply shocks** create a policy trade-off; **demand shocks** do not.
- The cost of disinflation depends on γ and on whether expectations are **anchored**.
- Denmark and the US since 2020 illustrate the dominant role of supply shocks plus anchoring.

Preview of Lecture 6

Next lecture you put AS and AD together to study the closed-economy AS-AD model:

- equilibrium in (y, π) space when both curves shift;
- dynamics of adjustment after demand and supply shocks;
- the central bank's optimal response and the Taylor principle revisited;
- how persistence in π^e shapes the impulse response.

The **conceptual bridge**: today's AS curve plus L4's AD curve give you the engine of all short-run macro analysis for the rest of the course.

Appendix: Derivation of the Acceleration Form

Start from the expectations-augmented Phillips curve:

$$\pi_t - \pi_t^e \;=\; -\alpha\,(u_t - \bar{u}) \;+\; s_t$$

Assume static expectations:

$$\pi_t^e \;=\; \pi_{t-1}$$

Substitute:

$$\pi_t - \pi_{t-1} \;=\; -\alpha\,(u_t - \bar{u}) \;+\; s_t \qquad \square$$

This is the **accelerationist** form: sustained $u_t < \bar{u}$ gives rising inflation. It is also the natural empirical specification for estimating α from a regression of $\Delta\pi_t$ on u_t.

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Appendix: Sacrifice Ratio Sketch

With anchored expectations $\pi^e = \pi^*$ and a one-off shock $s > 0$, the AS curve gives:

$$\pi_t - \pi^* \;=\; \gamma\,(y_t - \bar{y}) \;+\; s$$

To achieve $\pi_t = \pi^*$ on impact, the required output gap is:

$$\boxed{y_t - \bar{y} \;=\; -\frac{s}{\gamma}}$$

Implications:

- Larger shock s, larger output cost.
- Smaller slope γ (flatter Phillips curve), **larger** output cost: this is exactly why a flat curve makes disinflation expensive.
- Anchored expectations remove the additional cost from a shifting AS curve.

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Appendix: Where Does μ Come From?

The intercept μ in $\pi = \pi^e + \mu - \alpha u$ collects structural features of wage and price setting:

- desired markups and product market power;
- bargaining strength of workers and unions;
- the level and duration of unemployment benefits;
- matching frictions and search costs in the labour market;
- taxes on labour income.

Changes in μ shift $\bar{u} = \mu/\alpha$, that is, they move the natural rate. This is the channel through which **labour market reforms** affect long-run unemployment in the model.

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Appendix: Adaptive Expectations Updating

Adaptive expectations:

$$\pi_t^e = \pi_{t-1}^e + \eta\,(\pi_{t-1} - \pi_{t-1}^e), \qquad 0 < \eta \leq 1$$

- η small: expectations update slowly. Inflation has to deviate from π^e for a long time before π^e moves. This is the **anchored** limit when $\eta \to 0$.
- $\eta = 1$: expectations equal last period's inflation. This is the **static** case.
- Iterating: π_t^e becomes a weighted average of past inflation rates with geometrically declining weights.

The take-home: the speed at which past inflation feeds into current expectations determines how persistent inflation becomes after a shock.

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